Interfacing a 7-Segment Display with Arduino Uno

Using Wokwi Simulator — Tech Fest Project

7-Segment Display

Shows numbers 0–9

Arduino Uno

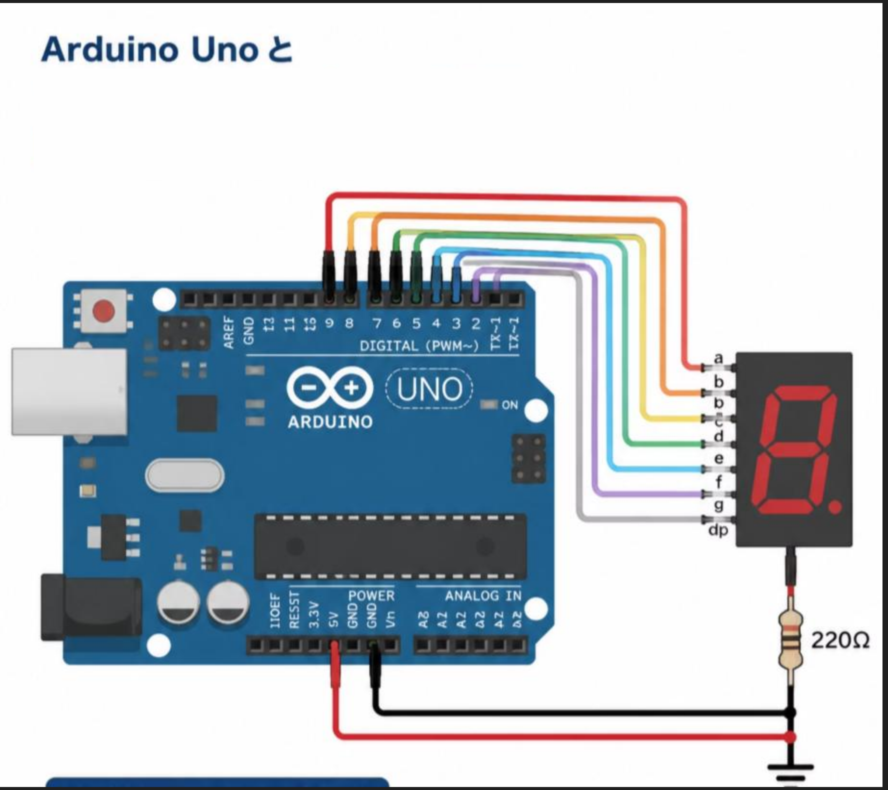
Controls the display

Wokwi Simulator

Test online — no hardware

Learn

Code, circuits, and outputs



What Is a 7-Segment Display?

A 7-segment display has **seven LEDs** arranged to show numbers. Each LED is called a **segment**.

Segment Labels

Segments **a** through **g** form numbers. **DP** is the decimal point.

Common Cathode

All cathodes join at **GND**. Send **HIGH** to light a segment.

Displaying Numbers

Lighting different segment combinations shows **0 through 9**.

Components Required

You need these six items for this project.

Project Setup
Use the items below to build and test the circuit in Wokwi.

Arduino Uno

Runs the program and controls the display.

7-Segment Display

Shows numbers 0 to 9.

Jumper Wires

Connect Arduino to the display.

USB Cable

Uploads code and powers Arduino.

Computer / Laptop

Runs the Wokwi simulator.

Wokwi Simulator

Tests the circuit online.

All these components work together to display numbers from 0 to 9 using Arduino Uno and the Wokwi Simulator.

Circuit Connections – Wiring Diagram

Arduino Uno

Single Common Cathode 7-Segment Display

D2→a

D3→b

D4→c

D5→d

D6→e

D7→f

D8→g

D9→DP

GND→COM

Circuit Connections – Pin Mapping

Segment	Arduino Pin	Function
a	D2	Top horizontal
b	D3	Top-right
c	D4	Bottom-right
d	D5	Bottom horizontal
e	D6	Bottom-left
f	D7	Top-left
g	D8	Middle
DP	D9	Decimal Point
COM	GND	Ground

Working Principle

1. Arduino
2. Digital Output
3. Segment ON
4. Number Displayed

Arduino sends HIGH and LOW signals to the segment pins. Different combinations of illuminated segments display digits from 0 to 9. The decimal point (DP) connected to D9 remains ON throughout the operation.

Program Logic

The Arduino code uses five key functions to control the display.

setup()

Runs once. Sets all pins as OUTPUT.

pinMode()

Defines each pin (D2–D9) as OUTPUT.

loop()

Runs repeatedly. Displays 0 to 9 in sequence.

digitalWrite()

Sends HIGH or LOW to each segment pin.

delay()

Pauses briefly so numbers are visible.

Output — Numbers 0 to 9

0 → 1 → 2 → 3 → 4 → 5 → 6 → 7 → 8 → 9

Observation

- Numbers change automatically from 0 to 9.
- The decimal point (DP) remains ON throughout the sequence.
- Each digit is displayed briefly before moving to the next number.

Real-World Applications

7-Segment Displays are widely used wherever numeric information needs to be displayed quickly and clearly.

- Digital Clock — Shows hours and minutes clearly.
- Calculator — Displays numbers and results.
- Scoreboard — Shows scores in sports events.
- Elevator Display — Shows the current floor number.
- Counter — Counts items on a production line.
- Fuel Pump — Shows litres and price.
- Digital Measuring Device — Shows voltage or current readings.
- Electronic Display — Used in timers and appliances.

Advantages of This Project

This project demonstrates a simple, low-cost, and effective method of learning digital display interfacing using Arduino.

Easy to Learn	Low Cost	Fast Response	Reliable
Low Power	Compact Size	Easy Programming	Widely Used

Conclusion

This project demonstrated how an Arduino Uno controls a Single Common Cathode 7-Segment Display. Using the Wokwi Simulator, we learned circuit connections, Arduino programming, and how digits 0–9 are displayed.

Arduino Basics

Display Interfacing

Wokwi Simulation

THANK YOU

Any Questions?